

T-740-SPMM Project D: Portfolio analysis

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Executive Summary

In this report, a portfolio strategy was established for the selection of projects. The strategy was defined with a long-term goal of maximizing profits and building long-term relationships with clients. Internal resources were defined and given a price baseline of 26.000 ISK per hour.

A prioritization model was created based on the following factors Return on Investment, Personnel capacity, Investment required, Necessary Compliance, Increase in Digital Processes and Probable Risk. Using those factors, priority was decided. The results were as follows.

1. Dala.family
2. Careflux
3. Ísland.is
4. Veiðiverfur

The first two projects will be started at once, and the third possibly soon after given strong personnel levels within the company. Given Personnel levels all projects could be worked on simultaneously but not at expected budget and man count, most cost planning failed to take into account the hidden expense of employment thus only calculating the paid salary not the cost of pension, union, vacation, and other fees.

Never the less, the current portfolio has met team expectations.

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Introduction

This report presents the team's work on Project D, focusing on the process of developing and applying a portfolio management strategy. This process is vital in real life, where portfolio selection needs to be carefully considered based on goals, resources and cost. The report begins by discussing the establishment of a clear portfolio strategy that defines the company's goals, available resources, and cost structure. The following section introduces the prioritization model, outlining the criteria, scoring system, and weighting method used to evaluate project proposals. Next, the report evaluates the four proposed projects based on these criteria, leading to a well-justified final project portfolio. The final section provides a reflection on the group's learning experience throughout this process, highlighting the challenges, decision-making steps, and outcomes of the team's collaboration.

Portfolio Strategy

The first step taken by the team was to establish a clear portfolio strategy. This strategy serves as the foundation for evaluating and selecting projects and is intended to align future decisions with the company's long-term vision. The goal of this discussion was to create a shared understanding of what the team wants the company to represent and to provide a consistent framework for guiding all project-related decisions. The team engaged in an open and collaborative discussion, which ultimately led to several important outcomes. The agreed-upon business goals are:

- To maximize profitability through effective project selection and delivery.
- To build long-term client relationships by emphasizing quality, reliability, and trust.

In addition, the team defined the available resources that will influence portfolio decisions:

- Internal resources: 1 UI/UX designer, 1 privacy and security compliance specialist, 4 backend developers, 4 full-stack developers, 8 frontend developers, 2 project managers, and 4 quality assurance testers.
- External resources: Subject-matter experts (e.g., legal consultants, compliance advisors, or other specialists) that can be engaged on an as-needed basis.

Finally, the team established the cost baseline for projects:

- Each internal or external specialist is valued at 26,000 ISK per hour. This rate will be used for estimating project costs and assessing profitability.

By consolidating these elements - business goals, internal and external resources, and cost structure - the team has defined a portfolio strategy that provides both direction and measurable criteria. This will enable the company to objectively evaluate incoming project proposals, prioritize them according to strategic fit and potential value, and select a balanced portfolio for the next six months.

Portfolio prioritization model

The next step for the team was developing the prioritization model, including the criteria, scoring system, and weights. Almost all criteria are evaluated on a scale from 1 to 5, where lower scores represent less favorable outcomes and higher scores indicate better performance. The only exception is the risk criterion, which is assessed separately on a scale from 1 to 16, ranging from no risk to high risk. Each criterion is assigned a weight that reflects its impact on the final score calculation.

The team chose the following criteria and weight:

- **Return on Investment (ROI):** It quantifies the anticipated financial return of each project. Lower values denote minimal or negative returns, while higher values indicate higher profit potential. ROI received the highest weight because financial return is the primary decision-driver for the team. A strong financial return also enables future investments, creating a positive reinforcement loop.
- **Personnel:** It evaluates the necessary level of team involvement. A lower score indicates a smaller required group, while a higher score indicates the need for more team members. Personnel effort is heavily weighted because the team has limited human resources. Projects that require fewer people or less specialized competence are more feasible and can be completed faster.
- **Investment:** It evaluates the financial contribution required from the organization. A higher score indicates minimal or no additional investment. Investment cost is important but weighted slightly lower than ROI, since a project may still be worthwhile even if it is expensive, as long as it returns more than it costs. The team does not want to automatically exclude high-investment projects, but it does want to differentiate between low-cost quick wins and capital-heavy initiatives.
- **Increasing Digital Services for Customers:** This evaluation measures the project's impact on customer-facing digital services. Lower categories are assigned to scores that reflect minimal or no improvement, while higher scores indicate progress toward digital transformation. This criterion supports long-term strategic goals like digital transformation and customer experience, but it is not a short-term priority compared to financial return. It is weighted lightly because while beneficial, it is not seen as immediately essential or revenue-critical.
- **Compliance Requirement:** This measure evaluates to what extent compliance is required. Compliance with regulations and laws costs money and takes time and work. A lower compliance requirement means that less effort is needed to reach compliance, whereas a higher compliance requirement might involve a more complex solution that requires external expertise to be compliant. Compliance is weighted highly because failing to meet legal

or regulatory requirements exposes the company to fines, delays, and repetitional damage. Compliance-heavy projects demand more oversight and external expertise, increasing cost and complexity.

- **Internal Digital Processes:** This evaluates the progress of internal processes brought by the projects. Higher scores indicate greater progress toward digital transformation. Improving internal processes provides operational value and efficiency, but similar to customer-facing digitalization, it has less short-term impact on financial performance.
- **Risk:** This is a metric that quantifies the level of unpredictability or potential problems associated with a project. A lower value indicates minimal risk, while a higher value denotes increased exposure. Risk is considered, but it has relatively low weight because risk can often be mitigated through planning, phasing, or external support. The team wants risk to influence prioritization, but not dominate it, to avoid being overly conservative and blocking innovation.

Once the model was defined, the team proceeded to individually evaluate the formal project proposals and assign scores to each criterion in the prioritization table. This process allowed them to calculate a total weighted score for each project, resulting in a clear ranking based on their overall strategic value and feasibility. The team came together and discussed their reasoning for the scoring they gave, and all agreed on a final model. The filled out prioritization models can be found in the Appendix.

Project Evaluation

The evaluation of the four project proposals was conducted to assess their relative value and alignment with the team's portfolio strategy. The project proposals were reviewed by each team member independently. Afterwards, the team convened and continued discussing and assessing the proposals together. Each project was reviewed based on a set of weighted criteria that reflect the team's business goals of maximizing profitability and fostering long-term client relationships. Each project proposal was scored based on the weighted criteria.

The result showed that Project 1 (Group 3 - Dala.family) achieved the highest overall score of 36.25, primarily due to its strong ROI, reasonable personnel requirements, and high scores in customer digital service, compliance, and internal process improvement. The project also presented a low overall risk relative to its projected return.

Project 3 (Group 5 - Careflux) followed closely with a total score of 32.75. It performed strongly in ROI, given that they have a variable funding focus. They also scored strongly in compliance and innovation-related criteria, reflecting its potential to advance the company's technical capabilities in AI-driven automation. The difference between the Careflux and Dala.family proposals lie in Careflux's slightly higher personnel requirements. This risk factor reduced their final weighted score compared to Dala.family.

Project 2 (Group 4 - Ísland.is) ranked third with a score of 23.25. The proposal scored highly primarily due to its strong compliance necessity and minimal investment needs. However, despite those advantages, its moderate ROI and limited impact on internal digital transformation resulted in a lower total score.

Lastly, Project 4 (Group 7 - Veiðivefur) received the lowest score of 17.75 due to its limited ROI potential, low compliance relevance, and minimal contribution to internal digital processes. The overall value resulted in this project scoring lower than the other projects.

In summary, the evaluation results were derived directly from the team's prioritization model. The ranking, based on scoring, is as follows:

1. Dala.family
2. Careflux
3. Ísland.is
4. Veiðivefur

These results form the team's foundation to justify their final portfolio selection, ensuring that the chosen sequence aligns with their strategic objectives and internal resource capacity.

Final project portfolio

Following the evaluation, the team selected a final project portfolio that aligns with its goals and portfolio strategy. The result of the weighted prioritization model formed the basis for the team's portfolio composition, with additional consideration given to project interdependence, duration, and personnel availability. The final selection order was determined as follows: Dala.family and Careflux will be executed simultaneously, followed by Ísland.is and lastly, Veiðivefur.

This order is based on the prioritization model scoring as well as sound reasoning from the team during their discussion meetings. Dala.family, which achieved the highest score (36.25), was chosen as the leading project because it offers a good return on investment, with manageable risks, and moderate personnel requirements. The team decided to start Careflux alongside Dala.family, recognizing the complementary goals and balanced use of resources between the two projects. Sharing external and internal expertise across both projects is expected to shorten the overall development time and improve efficiency. Careflux, which ranked second in scoring (32.75) in the prioritization model, carries slightly higher risks but also provides opportunities for creative and technical growth within the company. Running the two projects concurrently benefits the team due to the immediate financial benefits of Dala.family and the long-term intention for innovation from Careflux.

Given the team's strong staffing capacity, the Ísland.is project will be initiated once substantial progress has been made on the first two projects. This project ranks third in the prioritization model (23.25), and, while its ROI is lower, it provides substantial compliance and benefits. As a government-backed initiative, Ísland.is offers long-term strategic value by positioning the company as a trusted partner in the public sector and in the field of secure, public digital services. Starting this project after the initial two ensures that key specialists, particularly in UX, privacy, and compliance, are available when required.

Lastly, Veiðivefur, with the lowest score (17.75), will be completed last. With its low complexity, limited resource requirements, and minimal risk, the project is considered a suitable project once major developments from the higher-risk project are completed. This strategy primarily aims to maintain productivity and work continuity between larger projects.

The final portfolio order reflects a balanced approach that combines short-term financial gain with long-term strategic growth. The first two projects support one of the team's main business goals by generating income while increasing their innovation potential. In contrast, the third project enhances the team's credibility and reputation through public-sector collaboration, and the last project contributes to internal stability by serving as a lower-risk project, which concludes the phase of operation of this portfolio.

Conclusion

This report presented the development of a portfolio strategy and prioritization model for project selection within our company. The team established a clear strategic framework used to evaluate four projects across weighted criteria such as ROI, compliance requirement, risk, and contribution to digital transformation. The resulting prioritization led to the selection of Dala.family and Careflux as the first projects to be executed simultaneously, followed by Ísland.is and Veðivefur. This final portfolio reflects a balanced combination of short-term profit generation, innovation potential, and long-term strategic growth.

Reflecting on the process, project D presented several challenges for our team. We had to make assumptions about both budget and capacity and base our prioritization decisions on those estimations. This proved to be a complex and non-intuitive process that demanded extensive brainstorming and continuous reassessment of our priorities. What made it even more challenging was that none of us had prior experience with portfolio management. Nevertheless, through open discussions, collaboration, and a willingness to adjust our approach along the way, we eventually arrived at a portfolio we are satisfied with.

Prioritization Model

Measurement	ROI	Personnel	Investment	Increases Digital service to customers	Compliance - (must do)	Internal digital processes	Risk	Total Score	Prioritization
Weight	4	2	0,5	1	3	0,75	1		
Project 1 (Group 3)	4	3	5	4	5	1	8	36,25	1
Project 2 (Group 4)	2	3	5	3	5	1	12	23,25	3
Project 3 (Group 5)	5	2	5	4	5	3	15	32,75	1
Project 4 (Group 7)	1	5	5	3	0	3	4	17,75	4
Score (0-5)	1 = negative ROI 3 = break even 5 = positive ROI	1 = heavy involvement 3 = mild involvement 5 = light involvement	1 = exceeds budget 3 = within budget 5 = no investment	1 = does not increase service 3 = increases serves somewhat 5 = digital transformation	0 = does not apply 1 = is not needed due to compliance 3 = is good for compliance 5 = must do due to compliance (within timeframe)	1 = does not improve digital processes 3 = imports dp somewhat 5 = digital transformation	1 = no risk 8 = some risk 16 = high risk		

Figure A.1: Final Portfolio Prioritization Model

Measurement	ROI	Personnel	Investment	Increases Digital service to customers	Compliance - (must do)	Internal digital processes	Risk	Total Score	Prioritization
Weight	4	2	0.5	1	3	0.75	1		
Project 1 (Group 3)	4	2	5	5	5	1	7	36.25	1
Project 2 (Group 4)	3	3	5	2	5	1	14	24.25	3
Project 3 (Group 5)	4	2	4	4	5	3	16	27.25	1
Project 4 (Group 7)	2	5	5	3	1	4	5	24.5	4
Score (0-5)	1 = negative ROI 3 = break even 5 = positive ROI	1 = heavy involvement 3 = mild involvement 5 = light involvement	1 = exceeds budget 3 = within budget 5 = no investment	1 = does not increase service 3 = increases serves somewhat 5 = digital transformation	0 = does not apply 1 = is not needed due to compliance 3 = is good for compliance 5 = must do due to compliance (within timeframe)	1 = does not improve digital processes 3 = imports dp somewhat 5 = digital transformation	1 = no risk 8 = some risk 16 = high risk		

Figure A.2: Tinna's Portfolio Prioritization Model

Measurement	ROI	Personnel	Investment	Increases Digital service to customers	Compliance - (must do)	Internal digital processes	Risk	Total Score	Prioritization
Weight	4	2	0.5	1	3	0.75	1		
Project 1 (Group 3)	3	3	5	4	5	1	8	32.25	1
Project 2 (Group 4)	2	3	5	3	5	1	13	22.25	3
Project 3 (Group 5)	4	2	5	4	4	3	16	24.75	1
Project 4 (Group 7)	1	4	5	3	0	3	5	14.75	4
Score (0-5)	1 = negative ROI 3 = break even 5 = positive ROI	1 = heavy involvement 3 = mild involvement 5 = light involvement	1 = exceeds budget 3 = within budget 5 = no investment	1 = does not increase service 3 = increases services somewhat 5 = digital transformation	0 = does not apply 1 = is not needed due to compliance 3 = is good for compliance 5 = must do due to compliance (within timeframe)	1 = does not improve digital processes 3 = imports dp somewhat 5 = digital transformation	1 = no risk 8 = some risk 16 = high risk		

Figure A.3: Stella's Portfolio Prioritization Model

Measurement	ROI	Personnel	Investment	Increases Digital service to customers	Compliance - (must do)	Internal digital processes	Risk	Total Score	Prioritization
Weight	4	2	0.5	1	3	0.75	1		
Project 1 (Group 3)	4	3	5	5	5	1	7	38.25	1
Project 2 (Group 4)	2	4	5	3	5	1	12	25.25	3
Project 3 (Group 5)	5	3	5	4	5	3	14	35.75	1
Project 4 (Group 7)	2	5	5	2	0	3	4	20.75	4
Score (0-5)	1 = negative ROI 3 = break even 5 = positive ROI	1 = heavy involvement 3 = mild involvement 5 = light involvement	1 = exceeds budget 3 = within budget 5 = no investment	1 = does not increase service 3 = increases services somewhat 5 = digital transformation	0 = does not apply 1 = is not needed due to compliance 3 = is good for compliance 5 = must do due to compliance (within timeframe)	1 = does not improve digital processes 3 = imports dp somewhat 5 = digital transformation	1 = no risk 8 = some risk 16 = high risk		

Figure A.4: Julie's Portfolio Prioritization Model

Measurement	ROI	Personnel	Investment	Increases Digital service to customers	Compliance - (must do)	Internal digital processes	Risk	Total Score	Prioritization
Weight	4	2	0.5	1	3	0.75	1		
Project 1 (Group 3)	4	3	4	4	5	2	8	36.5	1
Project 2 (Group 4)	1	4	4	3	5	2	11	22.5	3
Project 3 (Group 5)	5	2	4	4	5	4	16	32	1
Project 4 (Group 7)	1	5	4	3	1	4	4	21	4
Score (0-5)	1 = negative ROI 3 = break even 5 = positive ROI	1 = heavy involvement 3 = mild involvement 5 = light involvement	1 = exceeds budget 3 = within budget 5 = no investment	1 = does not increase service 3 = increases services somewhat 5 = digital transformation	0 = does not apply 1 = is not needed due to compliance 3 = is good for compliance 5 = must do due to compliance (within timeframe)	1 = does not improve digital processes 3 = imports dp somewhat 5 = digital transformation	1 = no risk 8 = some risk 16 = high risk		

Figure A.5: Leonardo's Portfolio Prioritization Model

Measurement	ROI	Personnel	Investment	Increases Digital service to customers	Compliance - (must do)	Internal digital processes	Risk	Total Score	Prioritization
Weight	4	3	2	0.5	3	0.5	1		
Project 1 (Group 3)	4	3	5	4	5	1	8	44.5	1
Project 2 (Group 4)	2	3	5	3	5	1	12	32	3
Project 3 (Group 5)	5	2	5	4	5	3	15	39.5	1
Project 4 (Group 7)	1	5	5	3	0	3	4	28	4
Score (0-5)	1 = negative ROI 3 = break even 5 = positive ROI	1 = heavy involvement 3 = mild involvement 5 = light involvement	1 = exceeds budget 3 = within budget 5 = no investment	1 = does not increase service 3 = increases services somewhat 5 = digital transformation	0 = does not apply 1 = is not needed due to compliance 3 = is good for compliance 5 = must do due to compliance (within timeframe)	1 = does not improve digital processes 3 = imports dp somewhat 5 = digital transformation	1 = no risk 8 = some risk 16 = high risk		

Figure A.6: Kristján's Portfolio Prioritization Model